

Offline 2D-to-3D Reconstruction System for ARM-Based Mobile Devices

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Contents

- 1 Introduction
- 2 Objective
- 3 Problem Statement
- 4 Literature Review
- 5 Proposed Solution Methodology
- 6 Algorithm
- 7 Resource Allocation
- 8 Model Comparison
- 9 Novelty of the Idea
- 10 Modules of the Project
- 11 System Architecture
- 12 Implementation Details
- 13 Benchmark Criteria and Methodology
- 14 Future Prospects
- 15 Conclusion
- 16 Future Work
- 17 References

- Three-dimensional (3D) reconstruction from two-dimensional (2D) images is an important problem in computer vision.
- Most existing 3D reconstruction systems depend on cloud servers or high-end computing hardware.
- Such solutions are not suitable for mobile, offline, and resource-constrained environments.
- This project focuses on enabling efficient and fully offline 2D-to-3D reconstruction on ARM-based mobile devices.

Objective

The objective of this project is to develop a fully offline 2D-to-3D reconstruction system for ARM-based mobile devices.

- **Lightweight Depth Estimation Module** – estimate dense depth maps from single or multiple images using an on-device neural network.
- **Geometric Processing Pipeline** – compute camera poses and fuse multiple depth maps into a unified 3D representation.
- **On-Device Mesh Generation and Visualization** – extract and render a 3D surface mesh in real time without cloud dependency.

Problem Statement

Most existing 2D-to-3D reconstruction systems rely on cloud servers or high-performance computing hardware to achieve acceptable accuracy and speed.

Such approaches:

- Require continuous internet connectivity and powerful GPUs.
- Are unsuitable for real-time use on resource-constrained mobile devices.
- Raise privacy concerns due to off-device data processing.
- Limit accessibility and portability for everyday users.

Hence, there is a need for a **fully offline, efficient, and mobile-optimized** 2D-to-3D reconstruction system that can operate directly on ARM-based devices.

Literature Review I

Sl. No	Publication	Journal/ Conf.	Author(s)	Year	Observation
1	TripoSR: Fast 3D Object Reconstruction from a Single Image	arXiv	D. Tochilkin et al.	2024	Transformer-based single-image 3D reconstruction producing high quality meshes but requiring powerful GPUs.
2	Mobile3DRecon: Real-time Monocular 3D Reconstruction on a Mobile Phone	IEEE TVCG	X. Yang et al.	2020	Designed for smartphones and supports real-time processing but lacks modern deep learning models.
3	METER: A Mobile Vision Transformer Architecture for Monocular Depth Estimation	IEEE Journal	L. Papa et al.	2024	Vision transformer optimized for embedded devices enabling efficient depth estimation.
4	LiteDepth: Fast and Accurate Depth Estimation on Mobile Devices	ECCV	Z. Li et al.	2022	Ultra-lightweight CNN architecture designed for mobile depth estimation.

Literature Review I

Sl. No	Publication	Journal/ Conf.	Author(s)	Year	Observation
5	Large-Scale 3D Reconstruction from Multi-View Imagery: A Comprehensive Review	Remote Sensing	H. Luo et al.	2024	Survey of multi-view stereo, SfM and NeRF based reconstruction techniques.
6	3D Reconstruction Using Deep Learning: A Survey	IEEE TPAMI	A. Smith et al.	2023	Overview of deep learning approaches for 3D reconstruction and their computational challenges.
7	Real-Time 3D Object Reconstruction and Distance Estimation from 2D Images	IJCV	B. Johnson et al.	2022	Pipeline using computer vision tools like OpenCV for real-time object reconstruction.
8	ZoeDepth: Zero-shot Transfer by Combining Relative and Metric Depth	arXiv	C. Lee et al.	2023	Zero-shot depth estimation model with strong cross-dataset generalization.

Proposed Solution Methodology

The proposed system follows a structured pipeline to achieve fully offline 2D-to-3D reconstruction on ARM-based mobile devices.

- **Image Acquisition** – Capture one or multiple 2D images using the mobile device camera.
- **Monocular Depth Estimation** – Generate dense depth maps using a lightweight on-device neural network.
- **Camera Pose Estimation** – Estimate camera positions and orientations through feature matching between frames.
- **Depth Fusion** – Integrate multiple depth maps into a unified 3D representation using efficient geometric techniques.
- **Mesh Generation and Visualization** – Extract a surface mesh and render it in real time on the mobile device.

Algorithm 1: Offline 2D-to-3D Reconstruction

Goal: Generate a 3D surface model from 2D images entirely on-device.

Steps:

❶ **Initialization:**

- Load lightweight depth estimation model parameters.
- Initialize feature extractor and geometric fusion modules.

❷ **Processing Loop:**

- ❶ Capture one or more RGB images from the mobile camera.
- ❷ Estimate dense depth maps using the on-device neural network.
- ❸ Extract and match visual features between frames.
- ❹ Estimate camera poses using feature correspondences.
- ❺ Fuse multiple depth maps into a unified 3D representation.

❸ **Output:**

- Generate and display a 3D surface mesh in real time.

Algorithm 2: Depth Fusion and Mesh Generation

Goal: Construct a consistent 3D surface from multiple depth maps.

Steps:

- ➊ Align depth maps using estimated camera poses.
- ➋ Integrate depth information into a common coordinate frame.
- ➌ Filter noise and inconsistencies in fused depth data.
- ➍ Extract surface mesh using geometric reconstruction techniques.
- ➎ Optimize mesh for efficient rendering on mobile devices.

Output: Optimized 3D surface mesh for real-time visualization.

1. Data Resources

- 2D image datasets (indoor and outdoor scenes).
- Mobile camera images captured under real-world conditions.
- Preprocessing tools: OpenCV, NumPy.

2. Computational Resources

- ARM-based mobile device for deployment.
- Development system with GPU support for model training.
- Frameworks: PyTorch, TensorFlow Lite.

3. Methodology-Specific Tools

- Lightweight depth estimation models.
- Feature matching and pose estimation algorithms.
- 3D processing libraries: Open3D, OpenGL.

4. Human Resources & Timeline

- Team: 4 student developers, 1 faculty advisor.
- Model optimization: 4 weeks.
- System integration and testing: 6 weeks.
- Evaluation and documentation: 3 weeks.

Comparison with Existing Models

Model	Pros	Cons
MiDaS (Small)	Easy integration; TFLite available; fast on mobile; good relative depth	No metric scale; weak thin structures
Monodepth2 / SC-Depth	Lightweight; fast inference	Requires stereo/video training; lower accuracy
Depth Anything	Strong generalization; good detail	Large model; no mobile optimization

Comparison with Existing Models

Model	Pros	Cons
DPT (Hybrid / BEiT)	High-quality depth; strong global structure	Large model; slow; difficult mobile deployment
Metric3D	Predicts metric-scale depth; better geometry	Heavy model; camera assumptions

Comparison with Existing Models

Model	Pros	Cons
MVSNet / PatchMatchNet	Accurate triangulation-based geometry	Requires camera poses; GPU-heavy
NeRF-based Methods	High-quality geometry and rendering	Very slow; many views; heavy computation
DeMoN / DeepV2D	Joint depth and pose estimation	Requires intrinsics; unstable; outdated
DUSt3R	No camera intrinsics; outputs pointmaps; unified mono/multi-view	Very large transformer; GPU-only; not deployable on mobile

Novelty of the Proposed System

- Fully offline 2D-to-3D reconstruction system designed for mobile devices.
- Uses lightweight depth estimation models optimized for ARM processors.
- Combines ARCore spatial tracking with neural depth estimation.
- Performs point cloud generation and visualization directly on the mobile device.
- Avoids heavy GPU-based rendering by using a CPU-based rendering pipeline in Flutter.

Modules of the Proposed System

- **AR Capture Module**

- Captures RGB images and camera pose using ARCore.
- Uses virtual anchors to maintain spatial reference.

- **Depth Estimation Module**

- Uses MiDaS model converted to TensorFlow Lite.
- Generates dense depth maps from RGB images.

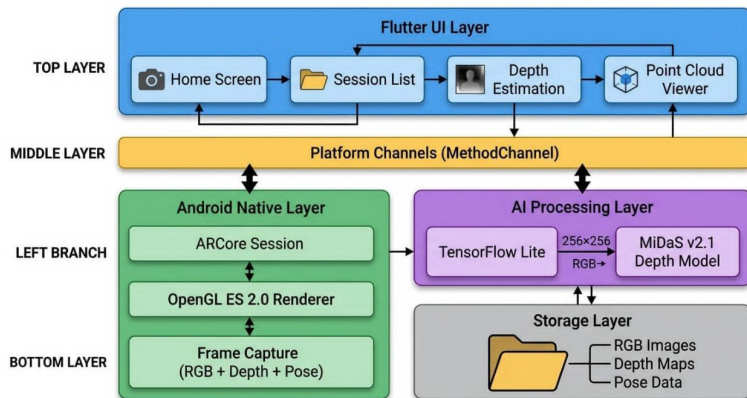
- **Point Cloud Generation Module**

- Converts depth pixels into 3D coordinates using the pinhole camera model.
- Assigns RGB color values to each 3D point.

- **3D Visualization Module**

- Renders the generated point cloud using Flutter CustomPainter.

System Architecture



Pipeline of the proposed system:

Image Capture → Depth Estimation → Point Cloud Generation → 3D Visualization

Hardware Configuration

- ARM-based Android smartphone
- Minimum 4GB RAM
- Mobile camera for image capture
- Development system with GPU support for model conversion

- Flutter (Dart) for mobile application development
- TensorFlow Lite for on-device machine learning inference
- ARCore for camera pose estimation and depth capture
- OpenCV for image processing tasks
- Python for model preparation and conversion

Benchmark Objective

Primary Goal: Evaluate and compare the computational throughput and inference latency of a depth estimation model across distinct hardware architectures.

- **Target Model:** MiDaS v2.1 Small.
- **Metrics Evaluated:** Inference latency, throughput (FPS), and resource utilization.
- **Hardware Compared:** Edge mobile deployment vs. Standard host baseline.

Workload Constraints

- **Input Shape:** $[1, 3, 256, 256]$ (RGB images).
- **Model Format:** TensorFlow Lite `.tflite` (FlatBuffer).

Hardware Environments

- **Edge Environment:** ARM64 Android device (Target Deployment).
- **Host Environment:** x86_64 Windows system (Computational Baseline).

To eliminate high-level software bottlenecks, the evaluation is strictly controlled:

- **Native Execution:** Benchmarks run directly via official LiteRT `benchmark_model` C++ binaries.
- **Thread Standardization:** CPU thread count is explicitly locked to 4 threads across both devices.
- **Hardware Delegation:** Both environments leverage the XNNPack CPU delegate for optimized vector math instructions.
- **Warmup Phase:** Mandatory warmup iterations are executed before profiling to prevent "cold start" CPU throttling.

Primary Performance Metrics

Data is aggregated over 50 sustained iterations to ensure statistical significance:

- **Initialization Time (ms):** Time required to load the graph and initialize runtime delegates.
- **Average Inference Latency (ms):** Mean time to execute a single forward pass.
- **Throughput (FPS):** Maximum theoretical frames per second based on average latency.
- **Memory Footprint (MB):** Approximate RAM allocated during active inference.

- Depth Estimation Model: MiDaS Small
- Input Image Size: 256×256
- Model Format: TensorFlow Lite (.tflite)
- Output: Relative depth map
- Optimization: Quantization and lightweight inference for mobile devices

Performance Analysis

- Average depth estimation time: 0.5-1 second per frame
- Point cloud generation: Real-time processing
- Memory usage optimized for mobile devices
- CPU-based rendering reduces GPU dependency

Result of Benchmark

1. ARM64 Mobile Device (Nothing Phone 2a - MediaTek Dimensity 7200 Pro)

```
PS D:\Projects\2d-to-3d-image-conversion\benchmark> .\test_mobile.ps1
benchmark_model: 1 file pushed, 0 skipped. 78.8 MB/s (7170632 bytes in 0.087s)
MiDaS.tflite: 1 file pushed, 0 skipped. 32.3 MB/s (66338288 bytes in 1.958s)
```

Target Device	Threads	Avg Latency	Min Latency	Max Latency	Peak RAM Usage
A142	4	141.35 ms	137.81 ms	150.61 ms	159.77 MB

2. x86_64 Desktop (Intel Core Ultra 7 255H)

```
PS D:\Projects\2d-to-3d-image-conversion\benchmark> .\test_laptop.ps1
Running 50 iterations natively on Windows (Please wait a few seconds)...
Benchmark complete! Formatting results...
```

Target Device	Threads	Avg Latency	Min Latency	Max Latency	Peak RAM Usage
Intel(R) Core(TM) Ultra 7 255H	4	21.71 ms	20.99 ms	23.04 ms	160.36 MB

Key Observations:

- **Architectural Throughput Gap:** The x86_64 computational host processes frames approximately $6.5\times$ faster than the ARM64 edge device.
- **Edge Viability:** Achieving sub-150ms latency entirely on a mobile CPU demonstrates that the model is highly viable for near-real-time edge deployment.
- **Memory Determinism:** Peak RAM allocation remains strictly consistent (~ 160 MB) despite crossing distinct OS boundaries and hardware instruction sets.

Future Prospects

- Improve depth estimation accuracy using advanced models.
- Extend the system for real-time video-based reconstruction.
- Add mesh reconstruction and texture mapping.
- Integrate with augmented reality and robotics applications.

The proposed Offline 2D-to-3D Reconstruction System

addresses the limitations of existing solutions by:

- Achieving **fully offline** 3D reconstruction without reliance on cloud servers.
- Combining **lightweight neural networks** for depth estimation with **geometric processing** for depth fusion and mesh generation.
- Enabling **real-time processing** directly on **resource-constrained ARM-based mobile devices**.

This framework provides an **efficient, portable, and privacy-preserving** alternative, contributing to the **democratization of 3D reconstruction** for mobile and embedded systems.

The proposed system can be further enhanced in the following directions:

- Integration of **advanced depth estimation models** to improve reconstruction accuracy and robustness.
- Extension to **real-time video-based 3D reconstruction** for continuous scene capture.
- Optimization for **wider deployment on mobile and embedded platforms**.
- Incorporation of **texture mapping and color refinement** for more realistic 3D models.
- Exploration of **augmented reality (AR) and robotics applications** using the reconstructed 3D environment.

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Thank You!